

**Lab Project**  
**EE 457 Fall 2024**  
**Lab #2**

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**For: Prof. Thuc**

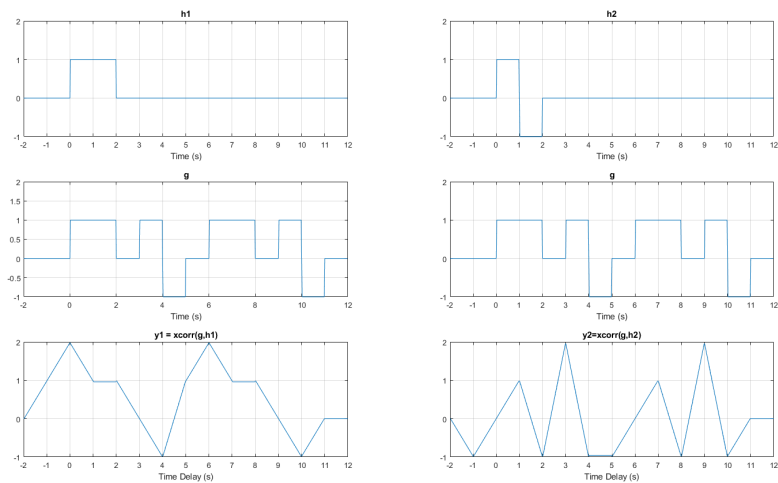
**Date: 11/15/24**

## Purpose of Description

The purpose of this lab is to understand the basic concepts of convolution and correlation and to analyze the energy and power of a discrete-time signal using MATLAB/Simulink

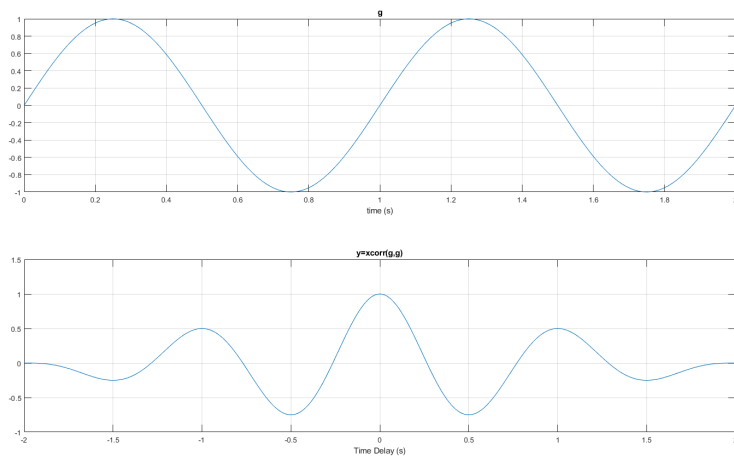
## Design and Implementation Procedure

First, I created the input signal  $g(t)$ , the rectangular pulse  $h_1(t)$ , and the piecewise pulse  $h_2(t)$ . I used MATLAB to calculate the cross-correlation between  $g(t)$  &  $h_1(t)$  and  $g(t)$  &  $h_2(t)$ .



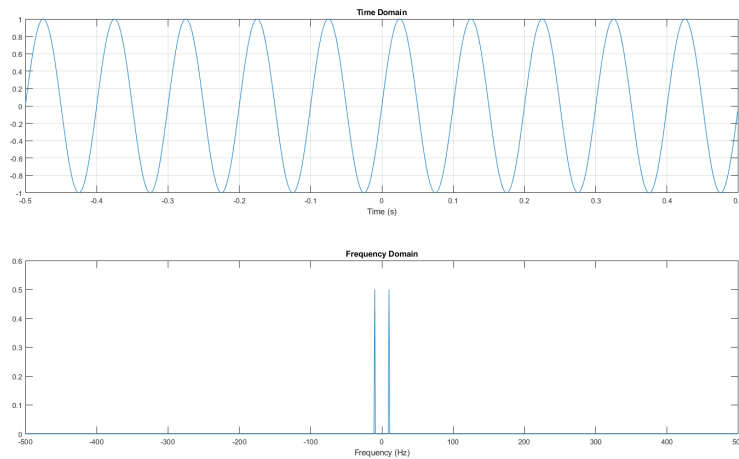
**Figure 1:** On the left is cross-correlation between signals  $g(t)$  and  $h_1(t)$ . On the right is the cross-correlation between signals  $g(t)$  and  $h_2(t)$ .

Next, I calculated the autocorrelation of a sine wave with a frequency of 1 Hz.



**Figure 2:** The autocorrelation of a sine wave with a frequency of 1 Hz.

After that, I calculated the Fourier Transform of a sine wave with the frequency of 10 Hz. Then, I calculated the total power of the sine wave from the time domain and from the frequency domain.

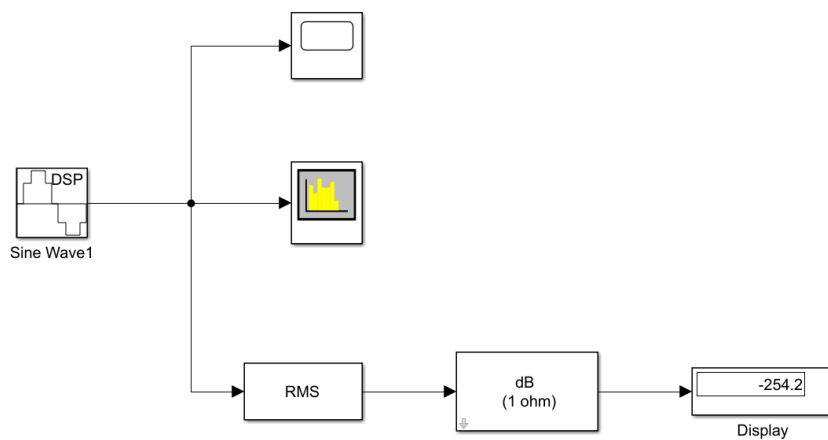


**Figure 3:** The Fourier Transform of a sine wave with a frequency of 10 Hz

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Total power of the signal g from time domain is 0.500000 (W) = 26.989700 (dBm)
Total power of the signal g from frequency domain is 0.500000
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**Figure 4:** The output of MATLAB that calculated the total power of the signal from the time domain and from the frequency domain.

Finally, I used Simulink to calculate the total power of a sine wave, along with visualizing the spectrum of the sine wave.



**Figure 5:** The simulink model calculates the total power of a sine wave, along with visualizing the spectrum of the sine wave.

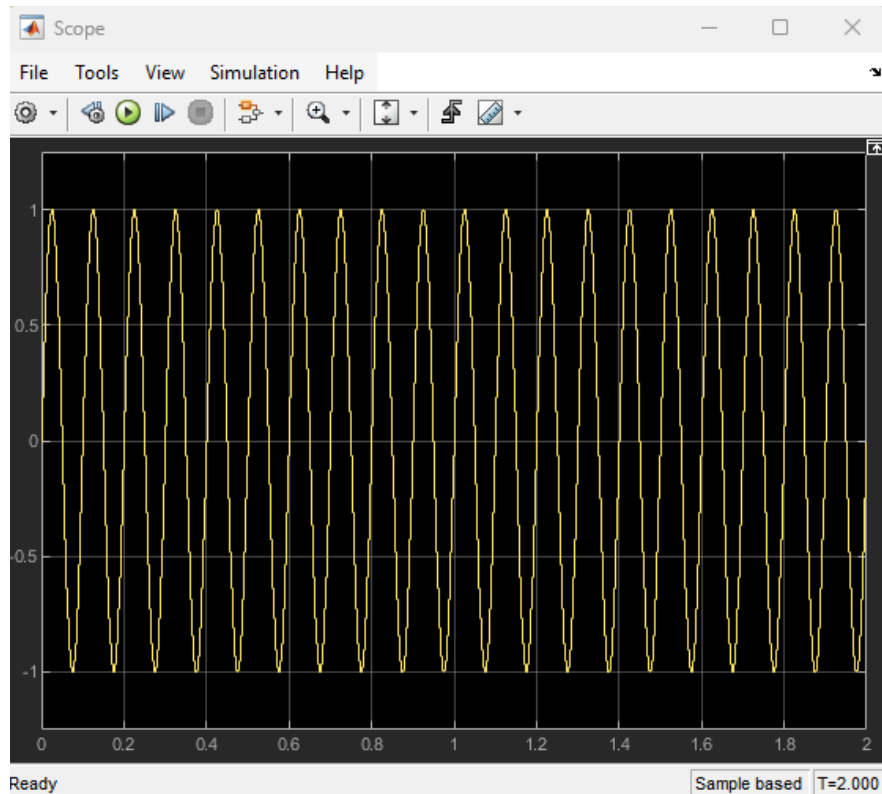


Figure 5: The sine wave signal in the time domain.

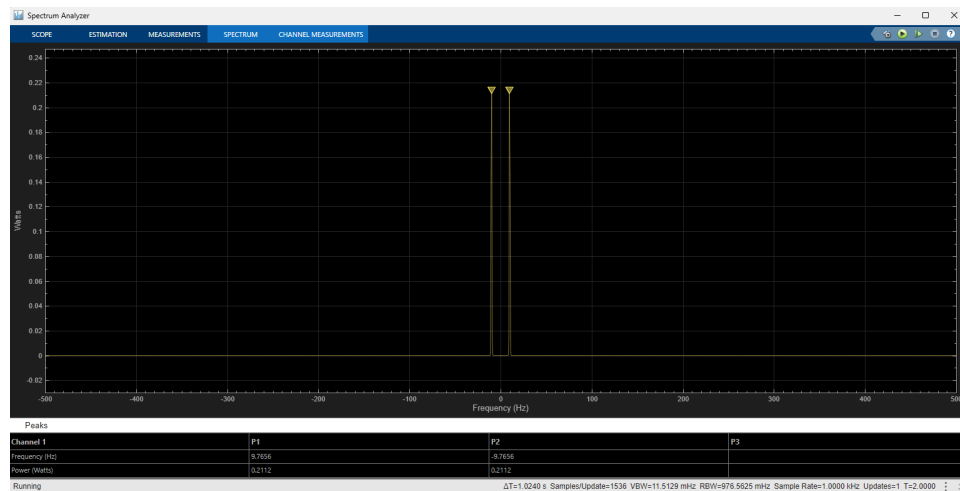


Figure 6: The spectrum of the sine wave signal.

In the output plots you can see the signal and spectrum of the signal are both correct. In the spectrum plot, the two peaks are at -10 and 10 Hz, which is to be expected. The power output is -254.2, which is not the same as the calculated power from the MATLAB code.

## Lessons Learned

Throughout this lab, I learned how to calculate the cross-correlation and autocorrelation of time signals. I learned how to calculate the total power of a signal from the time and frequency domain in MATLAB and Simulink.